

Lowkenuinely Non-Linear. Given n distinct integers $z_1, \dots, z_n$, we want to find an integer x that minimizes the following quantity:

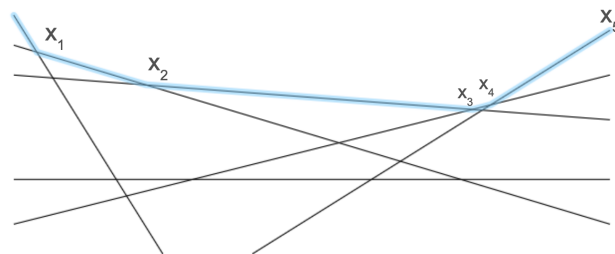
$$\sum_{i=1}^n (x - z_i)^2.$$

Give an $O(n)$ time algorithm for finding such a x and prove correctness of your algorithm.

Nobody Tell Kent Quanrud That I Stole Another One of His Problems. Let L be a set of n lines of the plane, where the i 'th line is of the form. $y_i(x) = a_i x + b_i$ for $a_i, b_i \in \mathbb{R}$. The *upper envelope* is the function

$$U_L(x) = \max_{i \in [n]} y_i(x).$$

Observe that U_L is piecewise-linear and convex. We represent the upper envelope as a list of m breakpoints $(x_1, \ell_1), \dots, (x_m, \ell_m)$ where $x_1 < \dots < x_m$ are the x -coordinates where the topmost line changes, and ℓ_i is the line achieving the maximum for $x \in [x_i, x_{i+1})$.



Design and analyze a recursive algorithm computing the upper envelope of n lines¹.

¹Hint: n lines induce $\binom{n}{2}$ points of intersection, but how many breakpoints can the upper envelope have?